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Course Orientation, Assessment, and Logistics

Your roadmap from business questions to an analysis portfolio

Week 1

Welcome: what this course is about

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Course purpose

Use data science to understand a life insurance retention problem and communicate a defensible business recommendation.

- Start with the stakeholder's decision, then choose the analysis.
- Work with clearly labeled synthetic policy and payment histories.
- Build a connected portfolio, rather than isolated exercises.

Today's goals

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By the end of orientation, you can:

1. Explain the course roadmap and final project expectations.
2. Describe how team work and individual contributions are assessed.
3. Identify your team, tools and preparation for Week 2.

Today's output

A team roster, a readiness checklist and any questions about assessment or logistics.

The 13.5-week roadmap

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Week 1	Course orientation, assessment and logistics.
Weeks 2–4	Understand the business problem, acquire data and prepare reliable histories.
Weeks 5–8	Analyze retention, explore warning signals, model and evaluate.
Weeks 9–10	Communicate findings and gather stakeholder acceptance feedback.
Weeks 11–13.5	Finalize the project; submit the final portfolio and reflection in Week 13.5.

The project: life insurance policy retention

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Fictional business scenario

A retention team has limited capacity for customer outreach. Which active policies should it prioritize for timely support, and on what evidence?

- Your instructor acts as the retention manager.
- Confirm the decision, definitions, capacity and constraints in Week 2.
- Findings demonstrate methods using synthetic data; they do not establish how real customers behave.

Three connected analyses

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1. **Retention performance:** how does policy lapse vary by product, channel and policy duration?
2. **Early warning patterns:** which earlier payment and service patterns are associated with subsequent lapse?
3. **Outreach prioritization:** does a simple classifier outperform an agreed overdue-payment rule?

For every analysis

State the business question, method, evidence, limitations and stakeholder implication.

What you will learn

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- Translate stakeholder needs into measurable analytical questions.
- Interpret insurance roles, products and policy events accurately.
- Prepare linked histories and analyze retention with valid denominators.
- Build and evaluate a simple model without using future information.
- Deliver reproducible evidence and customer-sensitive recommendations.

How the course works

Teams	3–5 students; each member records contributions.
Weeks 2–10	One 90-minute meeting: 8 minutes of insurance context, 12 minutes of data science and 70 minutes of project work.
Outside class	Approximately 1–2 hours per week.
Finalization	90-minute clinics in Weeks 11–13; an approximately 45-minute final checkpoint in Week 13.5.

How your work is graded

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Assessment component	Weight
Business requirements and problem framing	20%
Data preparation and reproducibility	20%
Analytical methods and evaluation	25%
Communication and recommendations	20%
Individual contribution and reflection	15%
Total	**100%**

The first four components are assessed at team level (85%). The final component is assessed individually (15%).

Grading evidence: framing and preparation

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Business requirements and problem framing: 20%

A clear outcome, decision and scope; stakeholder interview notes; requirements traceability; correct insurance roles and lifecycle definitions.

Data preparation and reproducibility: 20%

Valid histories, documented quality decisions, chronological splits, leakage controls and a workflow another person can run.

Grading evidence: analysis and communication

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Analytical methods and evaluation: 25%

Useful analyses, an appropriate business-rule baseline, chronological evaluation, and an honest interpretation of errors and limitations.

Communication and recommendations: 20%

A clear portfolio and insurance explanation; evidence-based, customer-sensitive actions; refresh instructions; a proposed pilot and monitoring plan.

Your individual assessment: 15%

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- Document your own contribution with concrete evidence.
- Explain one important decision and alternatives considered.
- Reflect on a limitation or mistake and what you learned.
- State what you would change next time and why.

Keep a contribution log

Record the task, your contribution, supporting file or artifact and the decision made. Update it throughout the course.

Quick quiz 1

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Question

Which components are assessed individually?

Discuss: is the entire portfolio graded separately for every student?

Answer and explanation

Quick quiz 1

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Question

Which components are assessed individually?

Discuss: is the entire portfolio graded separately for every student?

Answer and explanation

The first four components are team assessments totaling 85%. Individual contribution and reflection account for 15%. Every student supplies evidence of their own work.

Feedback, checkpoints and final submission

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- Weekly artifacts receive formative feedback and may be revised.
- Week 10: present a portfolio draft and record stakeholder feedback.
- Weeks 11–13: resolve feedback, check reproducibility and rehearse.
- Week 13.5: submit the final team portfolio and individual reflection.

Model performance is not the whole grade

A model that does not beat the baseline can earn full credit when evaluation is sound and the recommendation follows the evidence.

Quick quiz 2

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Question

Your model does not outperform the baseline. Should you hide that result?

Explain how the assessment criteria guide your response.

Answer and explanation

Quick quiz 2

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Question

Your model does not outperform the baseline. Should you hide that result?

Explain how the assessment criteria guide your response.

Answer and explanation

No. Report the result, verify the evaluation and recommend the approach supported by evidence. Sound methods and honest conclusions matter; a high score alone does not demonstrate valid work.

The final portfolio: evidence

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- One-page executive summary and business recommendation.
- Charter, labeled policy example, interview notes and requirements table.
- Dictionary, product assumptions, lifecycle, provenance and quality report.
- Three reproducible analyses and approximately six purposeful charts overall.
- Baseline comparison, held-out evaluation, error analysis and limitations.

The final portfolio: delivery

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- Final report or dashboard and a sample outreach prioritization table.
- Execution and refresh instructions.
- A proposed pilot and monitoring plan.
- Customer-support safeguards that respect needs, affordability and choice.
- Each student's contribution statement and individual reflection.

Before submission: use the final checklist in the syllabus.

Tools, teamwork and course logistics

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- Recommended tools: Python notebooks, pandas, plotting libraries and scikit-learn.
- Download the synthetic CSVs and data dictionary before Week 2. You will develop your own notebooks; solution code is not supplied.
- Agree team roles, shared files and a regular check-in; record everyone's contribution.
- Confirm calendar deadlines, submission platform and communication channel with the instructor.
- Confirm attendance, late-work, citation and AI-use policies with the instructor.

Your coding responsibility

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- Build your own reproducible notebooks using the supplied data dictionary and conceptual worked examples.
- Check that you can import a CSV, inspect columns, parse dates and summarize a small table.
- Keep raw files unchanged. Save cleaning decisions and use relative file paths.
- Bring tool or programming difficulties to the instructor early; each team must explain its own workflow.

Readiness check and preparation for Week 2

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Complete before the first topic discussion

1. Read the syllabus, grading rubric and final submission checklist.
2. Form a team of 3–5 and record your agreed roles.
3. Check that your tools work and confirm access to course materials.
4. Bring questions about the retention manager's decision and constraints.

Next week: business understanding and requirements. Interview the retention manager and draft your project charter.